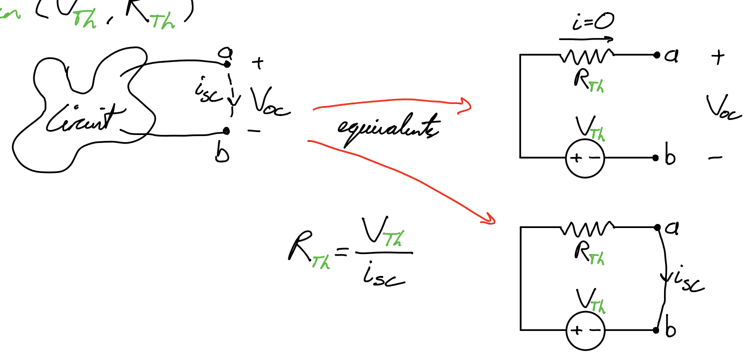
Thevenin and Norton Equivalents

- replace entire circuit, other than two nodes, with a resistor and voltage source in series
- replace entire circuit, other than two nodes, with a resistor and current source in parallel

- Elements
  - Open Circuit: $i_{oc} = 0 \rightarrow V_{oc}$
  - Short Circuit: $V_{sc} = 0 \rightarrow i_{sc}$

  - Ideal Ammeter: short circuit that displays current
  - Ideal Voltmeter: open circuit that displays voltage

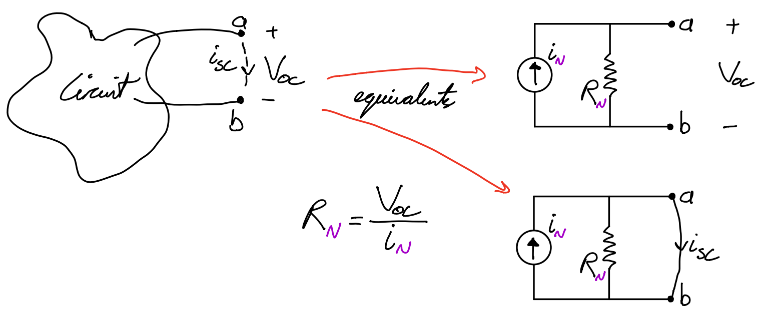
- Thevenin (V_{Th}, R_{Th})



$$R_{Th} = \frac{V_{Th}}{i_{sc}}$$

$$V_{oc} = V_{Th}$$

- Norton (I_N, R_N)



$$R_N = \frac{V_{oc}}{i_N}$$

$$i_{sc} = i_N$$

- if all sources are independent, kill them and calculate: $R_{eq} = R_{Th}$
- with dependent sources, calculate open circuit voltage (V_{oc}), and short circuit current (i_{sc})

P-Operator

- Can represent differential operator by a variable

$$3\frac{d^2y}{dt^2} + 5\frac{dy}{dt} - 8y = 14$$

$$3p^2y + 5py - 8y = 14 \rightarrow y = \frac{14}{3p^2+5p-8}$$

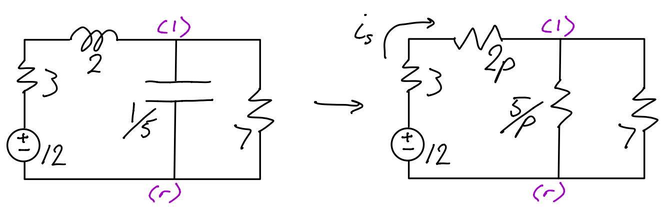
- Differentiate: $(f(x)=y)\cdot p$
- Integrate: $(f(x)=y) \div p$

- Apply to Inductor: $v = L\frac{di}{dt} = L\cdot pi$
- Apply to Capacitor: $i = C\frac{dv}{dt} \rightarrow v = \frac{1}{C}\cdot\frac{i}{p}$

- Impedance of Elements $(v = Zi)$

$$\left.\begin{aligned} &\cdot \text{Capacitor}: Z_C = 1/Cp \\ &\cdot \text{Inductor}: Z_L = Lp \\ &\cdot \text{Resistor}: Z_R = R \end{aligned}\right\} \text{All } Z \text{ behave like in Resistors}$$

Ex:



$$KI: \frac{12-V_1}{3+2p} = \frac{V_1\cdot P}{5} + \frac{V_1}{7}$$

$$\rightarrow V_1 = \frac{420}{14p^2+31p+50}$$

$$i_s = \frac{12-V_1}{3+2p} = \frac{84p+60}{14p^2+31p+50}$$

$$14p^2 i_s + 31pi_s + 50i_s = 84p + 60$$

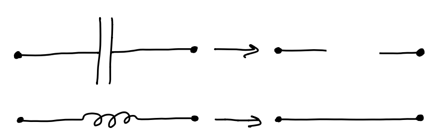
$$14i_s'' + 31i_s' + 50i_s = 60$$

Steady State

- Long term behaviour of circuits and elements

$$C\frac{dv}{dt} = i \quad \text{as } t\rightarrow\infty, \frac{dv}{dt}; \quad i = 0$$

$$L\frac{di}{dt} = v \quad \text{as } t\rightarrow\infty, \frac{di}{dt}; \quad v = 0$$

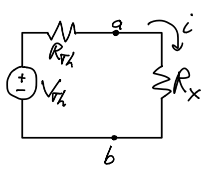


Energy in Circuit Elements

- Capacitor: $W = \frac{1}{2}CV^2$
- Inductor: $W = \frac{1}{2}Li^2$

Power of a Circuit

- Use Thevenin equivalent to power from a port

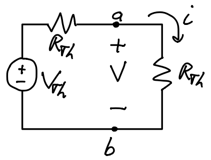


$$P_x = R_x\cdot i = \frac{R_x V_{Th}^2}{(R_{Th}+R_x)^2} \rightarrow \text{power from port a-b}$$

- Maximum power through port

$$\frac{dP_x}{dR_x} = \frac{d}{dR_x}\left(\frac{R_x V_{Th}^2}{(R_{Th}+R_x)^2}\right) = 0 \quad \text{(look for critical points)}$$

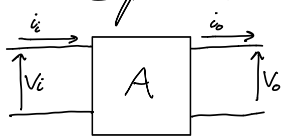
$R_x = R_{Th}$ (solving the differential equation)



$\therefore P_{max} = Vi$ $\qquad i = \frac{V_{Th}}{2R_{Th}}$, $V = \frac{V_{Th}}{2}$ (voltage divider)

$$= \frac{V_{Th}^2}{4R_{Th}}$$

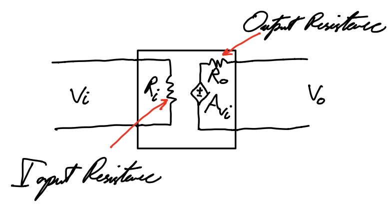
Operational Amplifiers



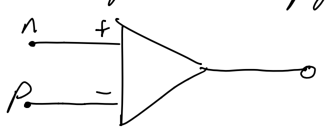
$A_V = \frac{V_o}{V_i}$ $\qquad A_V^{dB} = 20\log_{10}\left|\frac{V_o}{V_i}\right|$

$A_i = \frac{i_o}{i_i}$ $\qquad A_i^{dB} = 20\log_{10}\left|\frac{i_o}{i_i}\right|$

$A_P = \frac{P_o}{P_i}$ $\qquad A_P^{dB} = 10\log_{10}\left|\frac{P_o}{P_i}\right|$

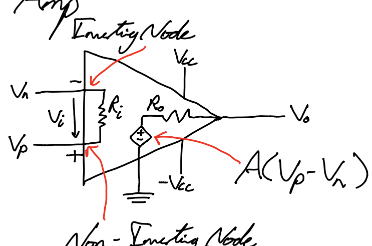


- Difference Amplifier
  - Takes 2 inputs and amplifies the difference



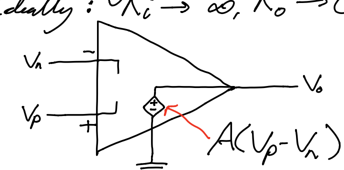
$A(V_p - V_n)$

- Op Amp



$R_i = 10^5 \rightarrow 10^{10}\ \Omega$
$R_o = 10 \rightarrow 10^2\ \Omega$
$A = 10^5 \rightarrow 10^8$ times bigger than $(V_p - V_n)$

- When doing MNA, do not create KCL equation for output node
- Ideally: $R_i \rightarrow \infty$, $R_o \rightarrow 0$, $A \rightarrow \infty$

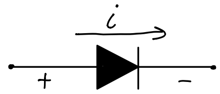


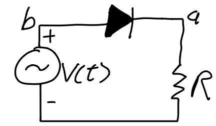
- Saturation
  - Output Voltage is bounded by: $-V_{cc} \leq V_o \leq V_{cc}$
  - If output voltage reaches extrema, OP Amp becomes saturated

$V_o = V_{cc} \rightarrow$ Positive Saturation
$V_o = -V_{cc} \rightarrow$ Negative Saturation

Diode

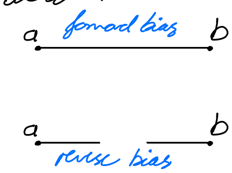
- Circuit Element that only allows current in one direction





$\xrightarrow{i>0}$ $v=0$

$\xrightarrow{v<0}$ $i=0$



- Current through Real Diode

$$i = I_S\left(e^{\frac{V}{nV_T}} - 1\right)$$

"−1" can be dropped for approximation

$$V_T = \frac{KT}{q}$$

Boltzmann's Constant

Temperature

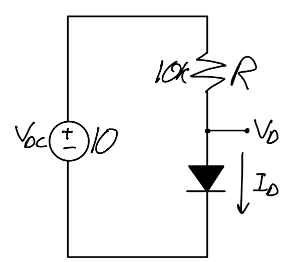
Electronic Charge

$\{1 < n < 2\}$ depending on type on Diode

- Iterative Solving Method
  - Compute I_S with test case:
    $(V_D, I_D) = (0.7, 0.001)$

$(V_D, I_D) = (0.7, 0.001)$

$n = 2, \; V_T = 0.025\,V$



$$i = I_S e^{\frac{V}{nV_T}} \longrightarrow I_S = \frac{0.001}{\exp\left(\frac{0.7}{2 \cdot 0.025}\right)} = 8.315E\text{-}10A$$

  - Two equations:

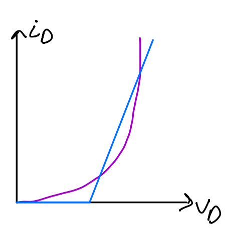
$$I_D = \underbrace{\frac{V_{DC} - V_D}{R}}_{1} = \underbrace{I_S\, e^{\frac{V_D}{nV_T}}}_{2}$$

  - Try a test case for V_D:
    $V_D = 0.7$
    1 $I_D = \dfrac{10 - 0.7}{10K} = 930\mu A$
  - Now use I_D to compute new V_D
    $I_D = 930\mu A$
    2 $V_D = (2)(0.025)\ln\dfrac{93E\text{-}5}{8.315E\text{-}10} = 0.6964V$

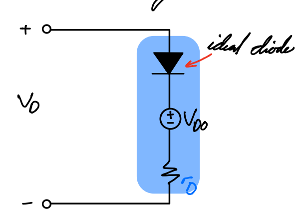
Repeat until Diode Voltage stops changing

$\therefore\; V_D = 696\,mV$
$\quad\;\; I_D = 930\,\mu A$

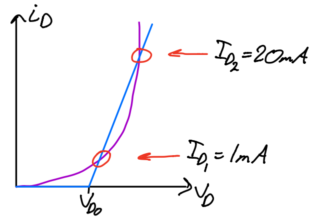
- Piecewise Model
  - Diode can be simulated with ideal diode, voltage source, and resistor



- Slope $= \dfrac{1}{r_D}$
- Exponential Characteristic of a Diode



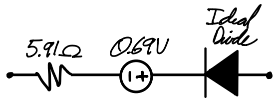
Ex: A diode has a voltage drop of 0.7V @ 1mA, $n=1.5$, $V_T=25mV$
Find piecewise model that is exact at 1mA & 20mA



$$I_S = \frac{I_{D_1}}{e^{V_{D_1}/nV_T}} = 7.82pA$$ current source

$$V_{D_2} = nV_T \cdot \ln\left(\frac{I_{D_2}}{I_S}\right) = 0.81V$$ voltage at I_{D_2}

$$\frac{1}{r_D} = \frac{I_{D_2} - I_{D_1}}{V_{D_2} - V_{D_1}} \Rightarrow 5.91\Omega$$ resistance of piecewise model



$$\frac{1}{r_D} = \frac{I_{D_2} - 0}{V_{D_2} - V_{D_0}} \longrightarrow V_{D_0} = 0.69V$$ voltage when current = 0

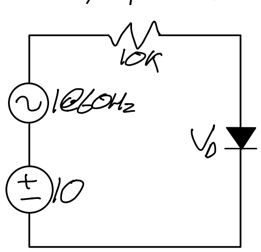
- Small Signal Model
  - Used when Diode operates with both AC and DC voltages
  - DC Steady State:
    - AC Voltage Source → short
    - AC Current Source → open
    - Inductor → short
    - Capacitor → open
  - AC Steady State:
    - DC Voltage Source → short
    - DC Current Source → open
    - Inductor → open
    - Capacitor → short
  - Notations
    - $\tilde{V}_a$ → AC voltage at (a)
    - $\overline{V}_a$ → AC voltage at (a)
    - $V_a = \tilde{V}_a + \overline{V}_a$

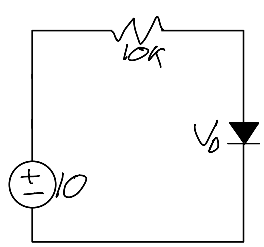
Ex:

$V_D = 0.7 @ 1mA$,
$n=2$, $V_T = 0.025V$

$$V_D = \tilde{V}_D + \overline{V}_D$$

$$I_S = I_D e^{-\frac{V_D}{nV_T}} = 8.32E\text{-}10A$$

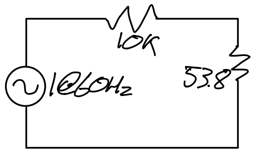




DC state

$\overline{I}_D = 0.93mA$
$\overline{V}_D = 0.69V$

$$r_D = \frac{nV_T}{\overline{I}_D} = 53.8\Omega$$

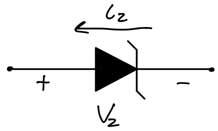


$$\tilde{V}_D = \frac{1}{10k + 53.8} \cdot 53.8 = 5.35\,mV$$

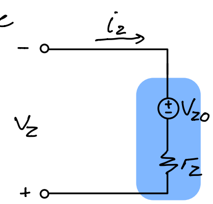
AC State

$$V_D = 0.69 + 5.35E\text{-}3\left(\cos\left(2\pi 60t + \varphi\right)\right)\ V$$

- Zener Diodes
  - Allows current to flow in reverse direction when voltage across terminals exceeds Zener Voltage



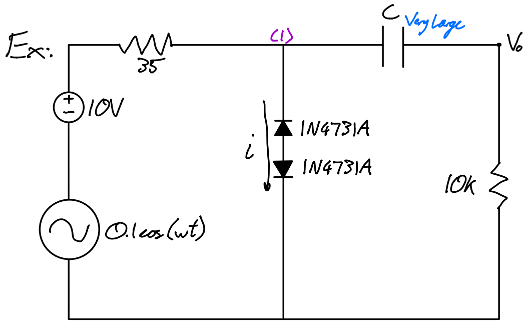
  - Can be represented using an incremental model
    - r_z is usually found from some test point (V_z, i_z)



$$\Delta V_z = \Delta i_z \cdot r_z$$

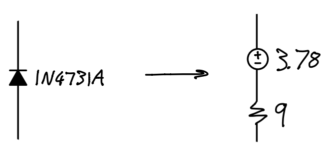
$$V_{z0} = V_z - r_z \cdot i_z$$

Ex:



From diode table: $(V_z, i_{zT}) = (4.3, 0.058)$

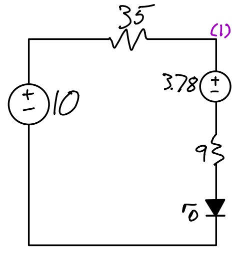
$r_z = 9\Omega$



Reverse Bias

$$V_{z0} = V_z - i_{zT} r_z = 3.78V$$

For circuit, zener diodes have $(V_F, I_F) = (0.7, 0.8E\text{-}3)$ when forward biased; $n=2$, $V_T = 0.025V$



DC State

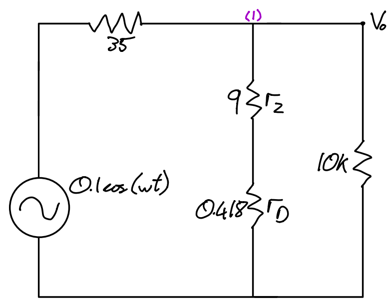
$$I_s = 0.8E\text{-}3 \cdot e^{-\frac{0.7}{0.05}} = 6.65E\text{-}10$$

$$e1:\ I_D = \frac{6.22 - V_D}{44}$$

$$e2:\ I_D = 6.65E\text{-}10 \cdot e^{\frac{V_D}{0.05}}$$

$\longrightarrow$ $V_D = 0.95V$, $I_D = 120mA$

$$r_D = \frac{nV_T}{I_D} = 0.418\Omega$$



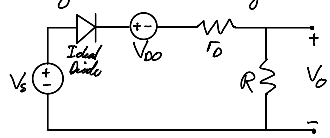
AC State

By Voltage Divider: $V_o = \frac{0.1\cos(\omega t)}{44.418} \cdot 9.418$

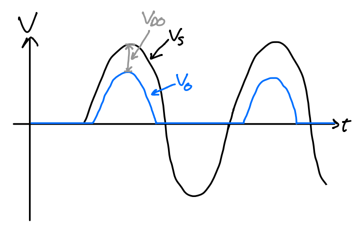
$= 0.0213\cos(\omega t)$

Rectifiers

- Can be made from Diodes
- Half wave Rectifier



$V_o(t) = \frac{R}{R + r_D}$



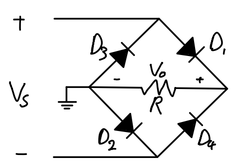
- Peak Inverse Voltage
  - The PIV is the maximum voltage before diode is at minimum

  Ex: for rectifier diode 1N4001, the PIV is 50V and they can dissipate up to 1W

  $$I = \frac{1}{50} = 20mA$$

  - For Half Wave Rectifier: $PIV = V_s$
  - For Full Wave Rectifier: $PIV = 2V_s - V_D$

- Bridge Rectifier
  - Functions as a full wave rectifier

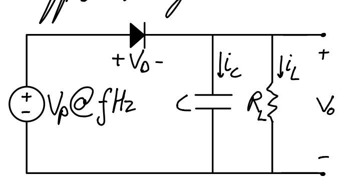


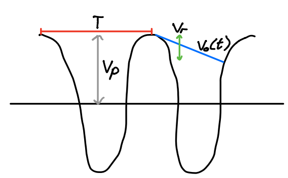
Diodes conduct in pairs: D1 & D2

D3 & D4

$PIV = V_s - V_D$

Ripple Voltage





- A decaying exponential resulting from a discharging capacitor: $V_o(t) = V_p \cdot e^{-\frac{t}{R_L C}}$

- After 1 period: $\begin{cases} V_p - V_r \doteq V_p \cdot e^{-\frac{T}{R_L C}} \\ e^{-\frac{T}{R_L C}} \doteq 1 - \frac{T}{R_L C} \end{cases} \rightarrow V_r = \frac{V_p - V_o}{f \cdot R_L \cdot C}$

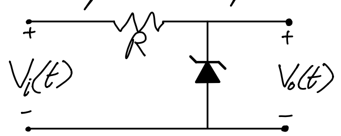
Half wave rectifier: $f = f$

Full wave rectifier: $f = 2f$

- Load Voltage: $V_{L(ave)} \doteq V_P - \frac{1}{2}V_r$
- Load Current: $i_L \doteq \frac{V_L}{R_L} = \frac{V_o}{R_L} \longrightarrow$ if ripple voltage is small: $i_L \doteq \frac{V_P}{R_L}$

Shunt Voltage Regulator

- The input and output will have DC and AC components:



$$V_i(t) = \bar{V}_i + \tilde{V}_i(t)$$
$$V_o(t) = \bar{V}_o + \tilde{V}_o(t)$$
$$\longrightarrow \quad \bar{V}_o = V_{Z0} + \frac{\bar{V}_i - V_{Z0}}{R + r_Z}\cdot r_Z \qquad \tilde{V}_o(t) = \frac{\tilde{V}_i(t)}{R + r_Z}\cdot r_Z$$

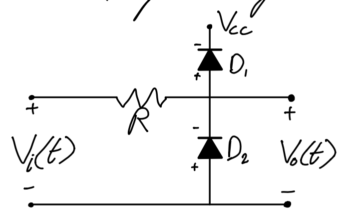
- In Shunt Regulators the output is affected by changes in the input voltage: (Line Regulation) and changes in the load current: (Load Regulation)

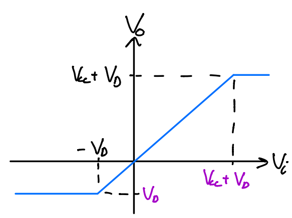
Line Regulation: $\frac{\Delta V_o}{\Delta V_i} \longrightarrow V_o = V_i \cdot \frac{r_Z}{r_Z + R}$

Load Regulation: $\left|\frac{\Delta V_o}{\Delta i_L}\right| \longrightarrow V_o = V_i \cdot \frac{r_Z \cdot R}{r_Z + R}$

Voltage Limiter

- Used to keep an integrated circuit voltage input between specified voltage margins



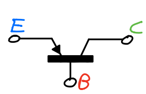


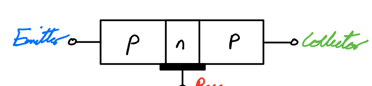
$$\{-V_D \le V_o \le V_{CC} + V_D\}$$

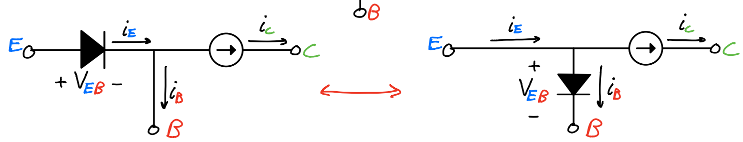
$$V_o = \frac{V_i}{R}\cdot R \longrightarrow \frac{\Delta V_o}{\Delta V_i} = \frac{R}{R} = 1$$

Bipolar Junction Transistors

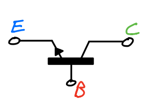
- PNP:

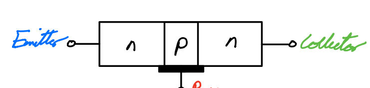


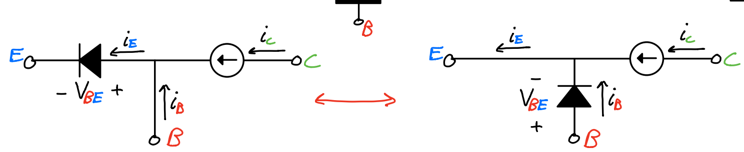




- NPN







- Relation Equations:

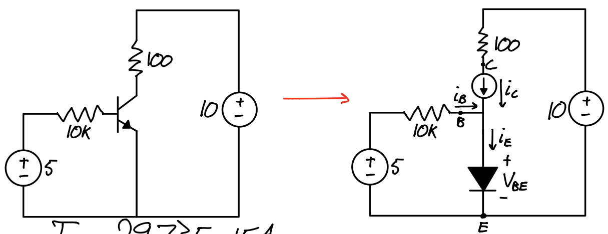
$$i_E = i_C + i_B = i_C\left(1+\frac{1}{\beta}\right) = i_B(1+\beta) = \frac{I_s \cdot e^{\frac{V_{BE}}{V_T}}}{\alpha}$$

$$i_C = \alpha \cdot i_E = \beta \cdot i_B = I_s \cdot e^{\frac{V_{BE}}{V_T}}$$

$$i_B = \frac{I_s \cdot e^{\frac{V_{BE}}{V_T}}}{\beta}$$

$$\alpha = \frac{\beta}{\beta+1} \qquad \beta = \frac{\alpha}{\alpha-1}$$

Ex: Calculate V_{BE} and i_C @ 293K



$I_S = 29.73E-15A$
$\beta = 100$

$V_B = V_{BE} + 0$

$0 = 5 - i_B \cdot 10K - V_{BE}$

$5 = \frac{I_s}{\beta} \cdot e^{\frac{V_{BE}}{V_T}} + V_{BE}$

$V_{BE} = 0.7V$
$i_C = 43mA$

- BJT Modes of Operation

	EB Junction	CB Junction	i_C
Cutoff — Opened Switch	Reversed Biased	Reversed Biased	$=0$
Active — Amplifier	Forward Biased	Reversed Biased	$=\beta \cdot i_B$
Saturation — Closed Switch	Forward Biased	Forward Biased	$<\beta \cdot i_B$

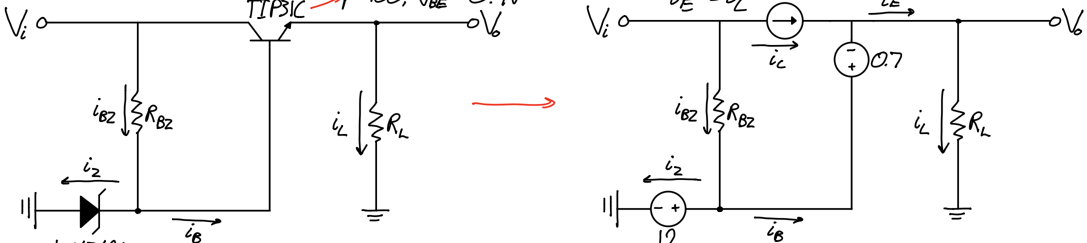
- DC BJT Circuit
  - Represent Diode from BJT with a Voltage Source

Ex: Voltage Regulator from a BJT and Zener Diode

Calculate R_{BZ} when $V_i = 15V$, $i_L = 1A$

TIP31C → $\beta = 100$, $V_{BE} = 0.7V$

$V_o = V_z - V_{BE} = 11.3V$

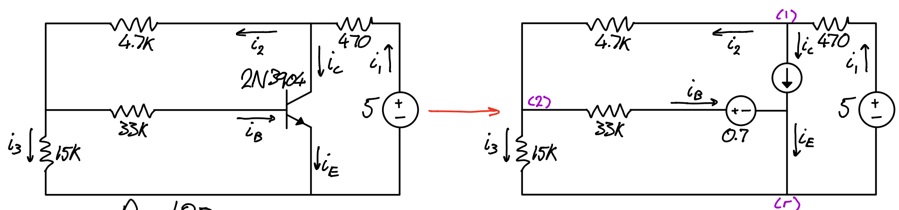


$V_z = 12V @ 21mA, r_z = 9\Omega$

$R_L = \frac{V_o}{i_L} = 11.3\Omega$ $\qquad R_{BZ} = \frac{V_i - V_z}{i_{BZ}} = 97\Omega$

$i_B = \frac{i_C}{\beta} = \frac{\alpha \cdot i_E}{\beta} = 9.9mA$ $\qquad i_{BZ} = i_B + i_z = 30.9mA$

Ex: Find i_B, i_C, i_E, V_{CE}



$\beta = 100$

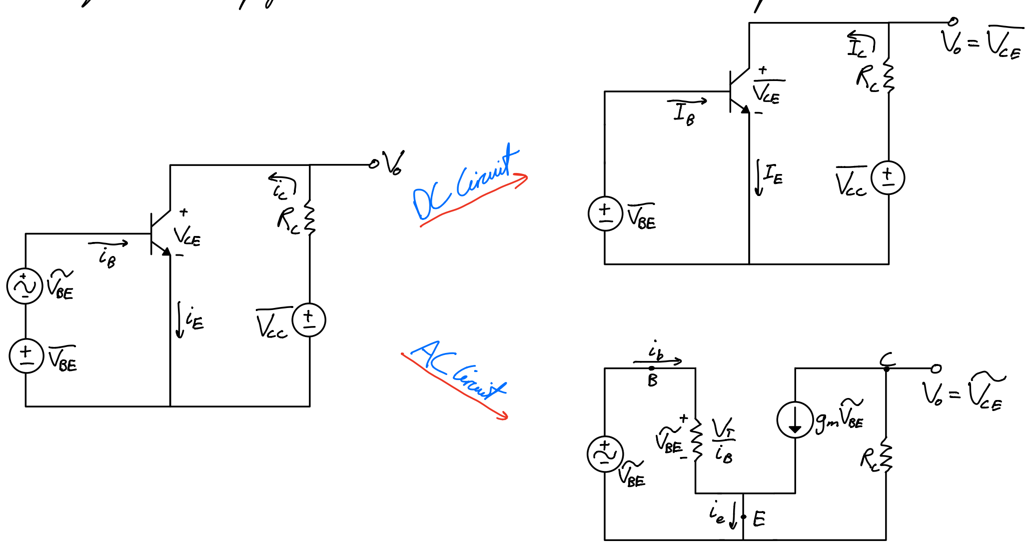
$i_B = 42\mu A$
$i_C = 4.20mA$
$i_E = 4.24mA$
$V_{CE} = 2.94V$

$$K1: \frac{5-V_1}{470} = i_C + \frac{V_1 - V_2}{4.7K} \qquad i_B = \frac{V_2 - 0.7}{33K}$$

$$K2: \frac{V_1 - V_2}{4.7K} = \frac{V_2 - 0.7}{33K} + \frac{V_2}{15K} \qquad i_C = 100 \cdot i_B$$

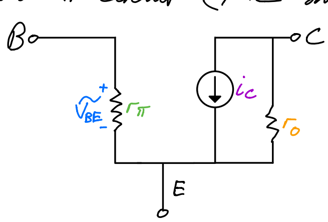
$$V_{CE} = V_1 - V_r \qquad i_E = 101 \cdot i_B$$

- BJT Amplifiers
  - Separate amplifier circuit into AC and DC components:



- This AC Circuit model fits the small signal AC equivalent of the Amplifier
- Solving BJT Amplifiers
  1) Solve DC circuit for I_C, r_π / r_e, g_m
  2) Solve AC circuit (Hybrid-π or T model) for $A_v = \frac{V_o}{V_i}$

- Hybrid-π Circuit (AC Small Signal Equivalent)



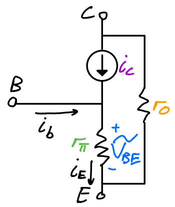
$$i_C = g_m \cdot \tilde{V}_{BE} = \frac{I_C}{V_T} \tilde{V}_{BE} = \frac{i_S \cdot e^{\frac{V_{BE}}{V_T}}}{V_T} \tilde{V}_{BE} = i_b \cdot \beta$$

$$r_\pi = \frac{\tilde{V}_{BE}}{i_b} = \frac{\beta}{g_m} = \frac{V_T}{I_B}$$

NPN: i_C, $\tilde{V}_{BE}$ PNP: $-i_C$, $-\tilde{V}_{BE}$

$$r_o = \frac{V_A}{I_C}, \text{ where } V_A \text{ is } V_{CE} \text{ when } i_C = 0$$

- T Circuit (AC Small Signal Equivalent)



$$i_c = g_m \cdot \tilde{V}_{BE} = \frac{I_C}{V_T} \tilde{V}_{BE} = \frac{i_s \cdot e^{\frac{V_{BE}}{V_T}}}{V_T} \tilde{V}_{BE}$$

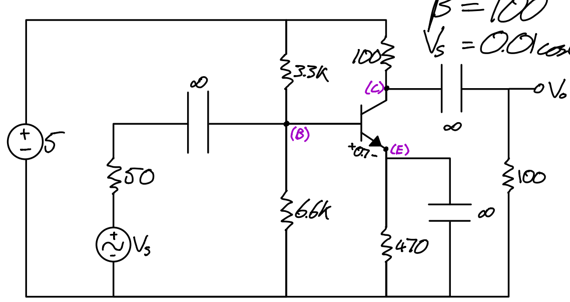
$$r_\pi = \frac{\alpha}{g_m} = \frac{V_T}{I_E}$$

$$r_o = \frac{V_A}{I_C}, \text{ where } V_A \text{ is } V_{CE} \text{ when } i_c = 0$$

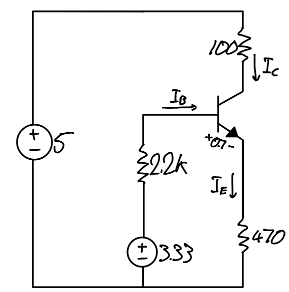
- Common Emitter Amplifier
  - In AC, the Emitter is connected to ground

Ex: Find $A_v = \frac{V_o}{V_s}$, V_c

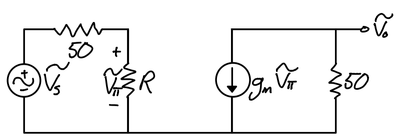
$\beta = 100$
$V_s = 0.01\cos(\omega t)$



DC



AC



$$3.33 = 2.2k \cdot I_B + 0.7 + (\beta+1) \cdot I_B \cdot 470$$
$$I_C = \beta \cdot I_B = 5.302mA$$

$$r_\pi = \beta \cdot \frac{I_C}{V_T} = 100 \cdot \frac{5.302E\text{-}3}{25E\text{-}3} = 476\Omega$$

$$R = (3.3k \| 6.6k \| 476) = 392\,\Omega$$

$$V_\pi = \frac{V_s}{50 + 392} \cdot 392 \qquad g_m V_\pi = \frac{I_c}{V_T} \cdot V_\pi = 0.21 V_\pi \text{ A}$$

$$V_o = -0.21 \cdot V_\pi \cdot 50$$

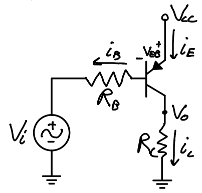
$$V_s = \frac{V_\pi}{0.887} \qquad A_v = \frac{V_o}{V_s} = -9.134$$

- Common Emitter Amplifier with Resistance in Emitter
  - In AC, the Emitter is connected to a resistance and then ground

- Common Base Amplifier
  - In AC, the Base is connected to ground

- BJT Switch — Saturation Operation
  - BJTs can operate in cut-off, saturation and active modes

Ex:



$V_{cc} = i_B R_B + V_{EB} + V_i$

if:

$V_i = V_{cc} \rightarrow i_B = 0 \longrightarrow$ cutoff

$V_i < (V_{cc} - V_{EB}) \longrightarrow i_B = \dfrac{V_{cc} - V_{EB}}{R_B} \longrightarrow$ active

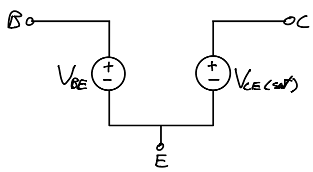
$V_o = i_c R_c$

$V_{EC} < V_{EC(saturated)} \longrightarrow$ saturation

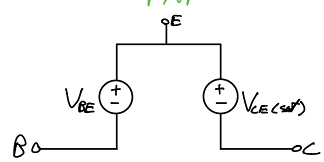
$V_o = V_{cc} - V_{EC(sat)}$

- A transistor operates as a switch in either cut-off or saturation modes
- In cut-off: transistor becomes open circuit
- In saturation: transistor becomes:

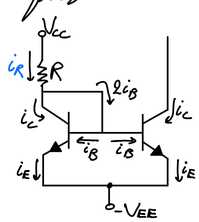
NPN



PNP



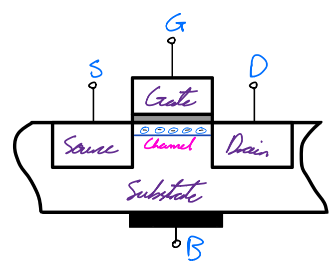
- Current Mirror
  - "Copies" current through one active device to another

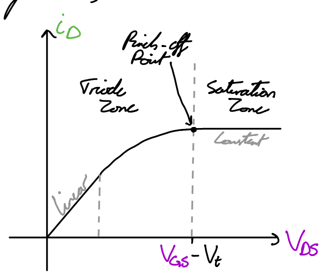


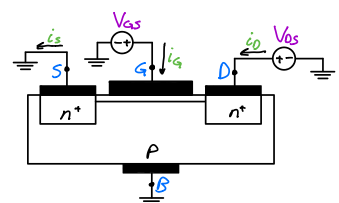
$i_c = i_R \cdot \dfrac{\beta}{\beta + 2}$

Mosfet

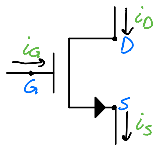
- When a large enough gate voltage is reached, the current will flow between the source and drain
- To create this channel, V_{GS} must be atleast the Threshold Voltage (V_t)







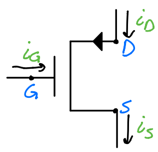
- NFET



  - Cutoff Region: $i_D = i_S = 0$, $V_{GS} < V_t$
  - Triode Region: $i_D = i_S = K_n' \cdot \frac{W}{L} \cdot \left[(V_{GS} - V_t) \cdot V_{DS} - \frac{V_{DS}^2}{2} \right] = \frac{V_{DS}}{r_{DS}}$

    $r_{DS} = \left[\mu_n C_{ox} \frac{W}{L} (V_{GS} - V_t) \right]^{-1}$, $0 < V_{DS} < V_{GS} - V_t$
  - Saturation Region: $i_D = i_S = \frac{1}{2} K_n' \cdot \frac{W}{L} \cdot (V_{GS} - V_t)^2 \left(1 + \frac{V_{DS}}{V_A}\right)$, $V_{DS} \geq V_{GS} - V_t$

- PFET



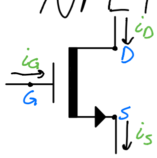
  - Cutoff Region: $i_D = i_S = 0$, $|V_{GS}| < |V_t|$
  - Triode Region: $i_D = i_S = K_p' \cdot \frac{W}{L} \cdot \left[|V_{GS} - V_t| \cdot |V_{DS}| - \frac{V_{DS}^2}{2} \right] = \frac{V_{DS}}{r_{DS}}$

    $r_{DS} = \left[\mu_n C_{ox} \frac{W}{L} (V_{GS} - V_t) \right]^{-1}$, $0 < |V_{DS}| < |V_{GS} - V_t|$
  - Saturation Region: $i_D = i_S = \frac{1}{2} K_p' \cdot \frac{W}{L} \cdot (V_{GS} - V_t)^2 \left(1 + \frac{|V_{DS}|}{|V_A|}\right)$, $|V_{DS}| \geq |V_{GS} - V_t|$

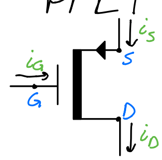
- Depletion MOSFET
  - Current between drain and source regardless of voltage at gate

    $V_t < 0$

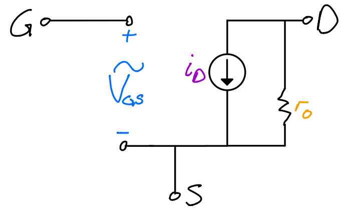
NFET Depletion



PFET Depletion



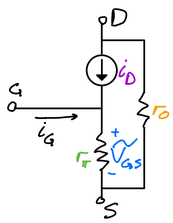
- Hybrid-π Circuit (AC Small Signal Equivalent)



$$i_D = g_m \cdot \tilde{V}_{GS} = K_n' \cdot \frac{W}{L} \cdot (V_{GS} - V_t) \cdot \tilde{V}_{GS}$$

$$r_o = |V_A| \cdot \left[\frac{1}{2} K_n' \cdot \frac{W}{L} \cdot (V_{GS} - V_t)^2\right]^{-1} = \frac{|V_A|}{I_D}$$

- T Circuit (AC Small Signal Equivalent)



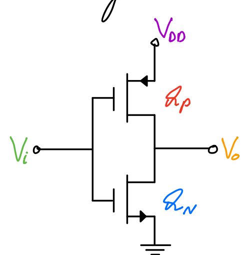
$$i_D = g_m \cdot \tilde{V}_{GS} = K_n' \cdot \frac{W}{L} \cdot (V_{GS} - V_t) \cdot \tilde{V}_{GS}$$

$$r_o = |V_A| \cdot \left[\frac{1}{2} K_n' \cdot \frac{W}{L} \cdot (V_{GS} - V_t)^2\right]^{-1} = \frac{|V_A|}{I_D}$$

$$r_\pi = \frac{1}{g_m} = \left[K_n' \cdot \frac{W}{L} \cdot (V_{GS} - V_t)\right]^{-1}$$

- Mosfets can be used in various amplifier circuits

- Complementary Mosfet (CMOS) Logic Inverter
  - An NFET and PFET where one is in Cut-off, and the other is in Triode
  - This arrangement makes $V_o = 0$ if $V_i = V_{DD}$, and $V_o = V_{DD}$ if $V_i = 0$



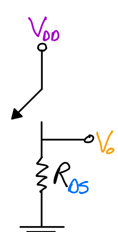
  - When $V_i = V_{DD}$:

$$V_{GS} = 0V \rightarrow \text{cutoff}$$

$$i_S = i_D = K_n' \cdot \frac{W}{L} \cdot \left[(V_{DD} - V_t) \cdot V_o - \frac{V_o^2}{2}\right]$$

Can approximate NFET to linear resistance: $R_{DS} = \left[K_n' \cdot \frac{W}{L} (V_{DD} - V_t)\right]^{-1}$

Simplifying the circuit to:



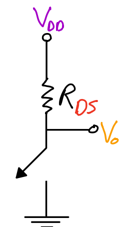
- When $V_i = 0$:

$$V_{GS} = 0V \rightarrow \text{cutoff}$$

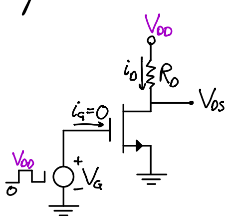
$$i_S = i_D = K_p' \cdot \frac{W}{L} \cdot \left[(|V_{DD}| - |V_t|) \cdot |V_o - V_{DD}| - \frac{(V_o - V_{DD})^2}{2} \right]$$

Can approximate PFET to linear resistance: $R_{DS} = \left[K_p' \cdot \frac{W}{L} (V_{DD} - |V_t|) \right]^{-1}$

Simplifying the circuit to:



- Mosfet Switches



V_G can be V_{DD} (Triode Mode) or 0 (Cut-off mode)

Can approximate NFET to linear resistance:

$$R_{DS} = \frac{V_{DS}}{I_D} = \left[K_n' \cdot \frac{W}{L} (V_{GS} - V_t) \right]^{-1}$$

Simplifying the circuit to: